

Technical Document #582: Data Engineering - Comprehensive Overview

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Data lineage tracks the origin and movement of data through systems. It is essential for debugging, compliance, and impact analysis.

ETL (Extract, Transform, Load) is a process that extracts data from source systems, transforms it to fit business rules, and loads it into a target data store.

Data lakes store raw data in its native format. They support structured, semi-structured, and unstructured data at any scale.

Data quality ensures data is accurate, complete, and consistent. Data validation, cleansing, and monitoring are essential components.

Data governance establishes policies and procedures for managing data assets. It covers data privacy, security, and compliance.

Augmented reality overlays digital information on the physical world. It has applications in training, maintenance, and entertainment.

Computer vision applications are becoming ubiquitous, from facial recognition to autonomous driving. Deep learning has enabled breakthroughs in visual understanding.

Quantum computing promises to solve problems that are intractable for classical computers. It has potential applications in cryptography, drug discovery, and optimization.

Cloud computing has democratized access to powerful computing resources. Startups can now access the same infrastructure as large enterprises.

Key Takeaways:

- ETL (Extract, Transform, Load) is a process that extracts data from source systems, transforms it to fit business rules, and loads it into a target data store.
- Data governance establishes policies and procedures for managing data assets.
- Stream processing handles continuous data streams in real-time.